CORE-Det: Computation Reallocation for Large-Scene SAR Target Detection

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***Abstract*—Large-scene synthetic aperture radar (SAR) target detection suffers from inefficient computation allocation because most processing is spent on background regions rather than target regions. We propose CORE-Det, a computation reallocation framework that dynamically redistributes processing across spatial regions, feature tokens, and training samples through a three-level sparsity-aware design tailored to SAR electromagnetic scattering. At the spatial level, the computation allocation via spatial distribution (COSD) module reconstructs a dynamic global spatial prior from sparse full-scene observations, performs patch-level screening before local detection, and fuses global and local confidence estimates to suppress background-generated target-like clutter. At the feature level, the computation allocation via feature value (COFV) module introduces SAR-aware mixture-of-experts routing, where specialist experts focus on target-dominant responses produced by strong scattering mechanisms, identity experts preserve structured contextual information at low cost, and zero experts suppress speckle-dominant and invalid activations. At the sample level, the decoupled error partitioning for adaptive retraining (DEPART) module mines hard cases caused by SAR scattering ambiguity and feeds module-specific error signals back to COSD and COFV, improving their ability to separate target-bearing regions from clutter and to distinguish target-dominant, contextual, and noise-dominant responses. Experiments on the self-built FUSAR-Airport and FUSAR-Harbor datasets with six baseline detectors show that CORE-Det consistently improves the accuracy-efficiency tradeoff. Compared with the corresponding baselines, it reduces GFLOPs by 88.92% and inference time by 51.97% on average, while improving mAP_{50} and recall by 9.59 and 8.94 percentage points (pp), respectively, and reducing false alarms by 19.61 pp.**

***Index Terms*—Computation-information alignment, large-scene sparse perception, mixture of experts, synthetic aperture radar (SAR).**

I. Introduction

SYNTHETIC aperture radar (SAR) provides all-weather, day–night imaging and has become essential for Earth observation applications such as maritime surveillance, military reconnaissance, disaster assessment, and infrastructure monitoring [1]. With continuing advances in satellite imaging, modern SAR systems now provide both wide swath coverage and high spatial resolution [2], [3], producing large volumes of imagery that require efficient interpretation. Although deep learning has substantially advanced SAR target detection, deploying these models on full-scene imagery remains difficult. Large-scene SAR data are extremely sparse, while on-orbit or near-real-time processing is constrained by limited computational resources. In addition, the coherent imaging mechanism of SAR causes complex backgrounds to generate target-like responses through multipath scattering, dihedral reflections, and speckle, which increases false alarms. As a result, existing detectors suffer from a severe mismatch between computational cost and useful scene information. As illustrated in Fig. 1, this mismatch appears in three coupled forms:

1) *Spatial Sparsity in Large Scenes:* Large-scene SAR images cover broad spatial extents, whereas targets occupy only a small fraction of the scene. Background structures such as buildings, roads, coastlines, and terrain corners can still produce strong responses through multiple scattering, dihedral reflections, and sidelobe effects, and these responses are further corrupted by speckle noise. Without global scene awareness, conventional detectors process vast homogeneous areas and clutter-prone regions almost uniformly, wasting computation and increasing the risk of clutter-induced false alarms.
2) *Feature Sparsity in Target Patches:* Even within retained target patches, only a small portion of features corresponds to truly discriminative scattering centers. The remaining features are often dominated by background clutter, weak contextual responses, or speckle-dominant noise. Existing detectors usually process these heterogeneous responses with nearly uniform computation, so low-value features continue to consume resources and interfere with target discrimination.
3) *Sample Sparsity in Training Data:* Training samples are likewise highly imbalanced in information value. Most samples are easy negatives from homogeneous backgrounds, whereas the truly informative cases are hard backgrounds that mimic targets and weak targets submerged in clutter. Uniform sampling therefore either wastes optimization on low-value regions or undertrains the model on confusing SAR-specific cases, leading to missed detections or false alarms.

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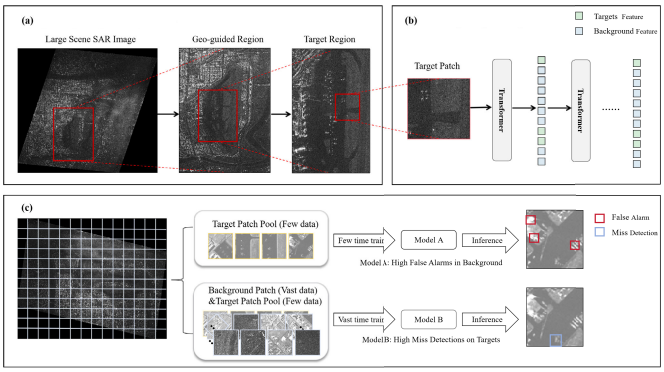


Fig. 1. Illustration of the inherent sparsity challenges in large-scene SAR object detection across three dimensions: (a) spatial sparsity in large scenes, (b) feature sparsity in target patches, and (c) sample sparsity in training data.

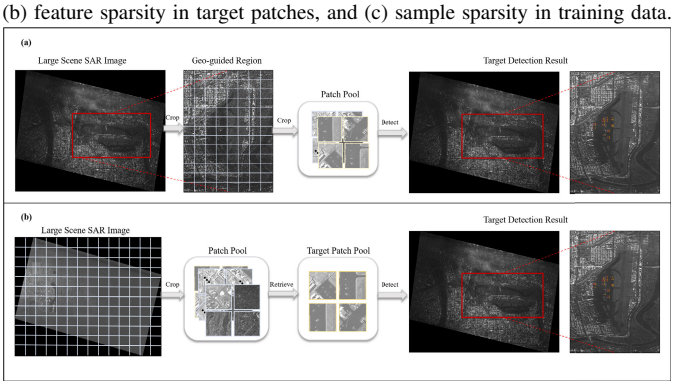


Fig. 2. Illustration of conventional large-scene SAR detection pipelines: (a) spatial-level processing via geo-guided region extraction and (b) patch-level processing via an additional classification module.

A. Prior Work in Large-Scene SAR Object Detection

Large-scene SAR object detection is fundamentally challenged by the extreme foreground–background imbalance in full-scene imagery. Since targets usually occupy only a tiny portion of the scene, exhaustive patch partitioning inevitably generates a large number of background-dominant patches, resulting in substantial redundant computation. To improve efficiency, existing methods typically introduce a preliminary filtering stage before full detection. From the perspective of where redundancy is removed, prior work can be broadly grouped into two paradigms: spatial-level processing via geo-guided region extraction and patch-level processing via an auxiliary patch discriminator. The two conventional large-scene SAR detection pipelines are illustrated in Fig. 2.

For spatial-level processing, traditional pipelines usually rely on external and largely static geographic or scene priors to reduce the searchable area, after which only the retained subregions are partitioned into patches for downstream detection. This strategy is particularly suitable for structured SAR scenes with strong environmental regularity, such as airports and maritime areas. For instance, Luo et al. [4] introduced geospatial contextual priors for panoramic SAR aircraft detection, where approximate airport location, geographic information matching, and optical-to-SAR registration are jointly utilized to localize airport regions before fine-grained aircraft detection. In maritime scenarios, similar ideas have been explored through sea-land segmentation and coastal-region constraints. Cui et al. [5] introduced a coarse-grained sea-land segmentation module to suppress false alarms from land regions, while Jia et al. [6] further proposed a coarse-to-fine three-stage framework, in which land-sea segmentation and CFAR are first used for fast candidate retrieval, followed by scene-level ship detection and target-level false-alarm discrimination. Although these methods can effectively shrink the search range, the retained regions are still relatively coarse, and because the priors are largely predefined rather than generated from the current SAR content, they cannot adaptively reflect scene-dependent scattering changes or target motion.

For patch-level processing, the full scene is first densely partitioned into patches, after which a lightweight discriminator is introduced to determine whether each patch should be forwarded to the detector. Existing methods can be further divided into decoupled prediscrimination and backbone-integrated discrimination. In the former, a coarse filtering stage is applied before candidate patches enter the detector. Chen et al. [7] designed a cascaded framework with two fully convolutional networks, where the first rapidly retrieves suspicious regions from large-scale imagery and the second performs refined localization. Zhang et al. [8] proposed MLBR-YOLOX, which incorporates a plug-and-play spatial patch detector to prelocalize potential target regions. Alternatively, patch discrimination can be embedded within the detection network itself. For example, $\mathcal{R}^2$-CNN [9] introduces a lightweight classifier on top of the backbone to suppress background patches, while subsequent works [10], [11] extend this idea by incorporating objectness prediction modules into intermediate network stages for end-to-end patch filtering. Despite their effectiveness, these methods operate only after dense patch partitioning and therefore do not exploit geographic priors to constrain patch generation itself. However, even after patch retention, most detectors treat all spatial features within a patch uniformly, despite the fact that target responses typically occupy only a small fraction of the region. As a result, background-dominated responses continue to dominate feature computation, leading to redundant processing and reduced discriminability.

Overall, existing methods either rely on coarse scene-level pruning or perform local patch-level discrimination after dense patch partitioning. The former can reduce the search space but still preserves many invalid patches, whereas the latter does not exploit geographic priors to constrain the search space from the source. More fundamentally, both paradigms remain patch-level perception schemes and therefore lack holistic awareness of the global scene. In sparse SAR imagery, such localized modeling cannot fully leverage global spatial priors to suppress target-like clutter, making it difficult to jointly utilize environmental constraints and local semantic evidence for robust decision-making. In addition, current approaches mainly focus on coarse patch-level redundancy reduction, while paying insufficient attention to fine-grained feature-level computation allocation. As strong-scattering background structures can easily produce target-like responses in SAR imagery, even retained target patches often contain a large proportion of low-value or interfering features, leading to unnecessary computation and degraded discriminability. Beyond inference, existing training strategies also tend to treat all samples uniformly, without adequately considering their heterogeneous information value. Therefore, an effective large-scene SAR

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detector should dynamically infer scene-dependent spatial priors and integrate global contextual constraints with local semantic evidence through a unified confidence mechanism. It should further perform feature-aware computation allocation to suppress low-value responses and prioritize informative samples during optimization.

B. Motivation

These limitations stem from the lack of explicit sparsity awareness across the scene, feature, and sample levels. A more effective detector should therefore allocate computation according to information value rather than treating all regions, features, and samples equally. To this end, we propose CORE-Det, a computation reallocation framework that addresses spatial, feature, and sample sparsity in a unified manner. The main contributions are as follows.

1) A sparsity-aware computation reallocation framework, termed CORE-Det, is presented for large-scene SAR target detection, which dynamically redistributes computational resources across spatial regions, feature tokens, and training samples. Tailored to the complex electromagnetic scattering environment of SAR imagery, this framework enhances the alignment between scene information content and computational cost.
2) A computation allocation via spatial distribution (COSD) module is developed to generate dynamic global spatial priors from sparse full-scene observations. Through continuous global spatial prior reconstruction and patch-level screening prior to local detection, COSD effectively suppresses low-value background regions and clutter induced by complex background scattering. In addition, a joint confidence mechanism is designed to combine global spatial prior confidence with local detection confidence, enabling collaborative decision-making between global environmental constraints and local semantic evidence for false alarm reduction.
3) Driven by feature value, a computation allocation via feature value (COFV) mechanism is constructed based on a ternary mixture-of-experts design that matches SAR scattering characteristics. Specifically, COFV routes target-dominant features from strong coherent scattering and dihedral reflections to specialist experts, retains structured contextual scattering via identity experts, and suppresses coherent speckle noise and homogeneous clutter through zero experts, realizing fine-grained computation allocation at the feature level.
4) To further boost sample efficiency, a decoupled error partitioning for adaptive retraining (DEPART) module is introduced to divide inference failures into module-specific hard cases and provide targeted feedback for the retraining of COSD and COFV. This design enhances the capability of the two modules in distinguishing real target scattering from confusing background scattering patterns, and introduces no extra computational overhead during the inference phase.

II. Related Work

A. SAR Target Detection

SAR target detection has evolved from classical signal-processing methods to deep detectors and, more recently, to efficiency-oriented designs for full-scene imagery. Early methods mainly relied on CFAR and handcrafted geometric or texture features [12], [13], [14], [15], which are efficient but limited in complex scattering environments. Modern single-stage and two-stage detectors substantially improve representation capacity through stronger backbones, oriented bounding boxes, and context modeling [16], [17], [18], [19], [20], [21], [22], [23], [24]. However, when these detectors are applied to gigapixel SAR scenes with sparse targets, exhaustive sliding-window inference becomes computationally prohibitive because most computation is still spent on background regions [25]. To reduce this redundancy, recent large-scene SAR methods mainly operate in two ways. One line introduces lightweight patch-level filtering, such as cascaded fully convolutional networks [7], standalone patch detectors like MLBR-YOLOX [8], and backbone-integrated objectness modules such as $\mathcal{R}^2$-CNN [9], OAN [10], and OPD-Swin-Transformer [11]. Another line incorporates scene-specific priors, including sea-land segmentation [5] and CFAR-guided coarse-to-fine filtering [6], to suppress clutter before fine detection. These methods reduce spatial redundancy at the patch or scene level, yet none jointly addresses global spatial priors, feature-level computational allocation, and hard-sample optimization within a unified SAR-oriented framework, leaving a critical gap that motivates the present work.

B. Dynamic Computation and Mixture of Experts

The mixture-of-experts paradigm routes inputs to specialized subnetworks, decoupling model capacity from inference cost and constituting a foundational mechanism for input-adaptive computation. Soft MoE [26] overcomes the training instability and token-dropping issues of sparse MoE through implicit soft assignment. Mixture-of-depths [27] constrains the number of tokens participating in computation via a top-K mechanism, enabling dynamic floating-point operations (FLOPs) allocation along the depth dimension. Zamfir et al. [28] introduce "complexity experts" that route simple tasks to low-cost experts, effectively reducing redundant computational overhead. In the remote sensing and SAR domains, MoE applications have primarily focused on cross-modal fusion and multitask adaptation. RingMoE [29] constructs a hierarchical architecture comprising modality-specific and shared experts to achieve unified representation across optical, multispectral, and SAR data. SM3Det [30] introduces grid-level sparse MoE to address feature discrepancies in multimodal remote sensing detection. However, existing methods leverage expert systems predominantly to decouple modality or task conflicts, leaving the extreme feature-space sparsity inherent to large-scene SAR detection largely unaddressed. More importantly, they are not explicitly designed for SAR-specific scattering behavior, where a few target-dominant responses produced by coherent scattering and dihedral reflections coexist with

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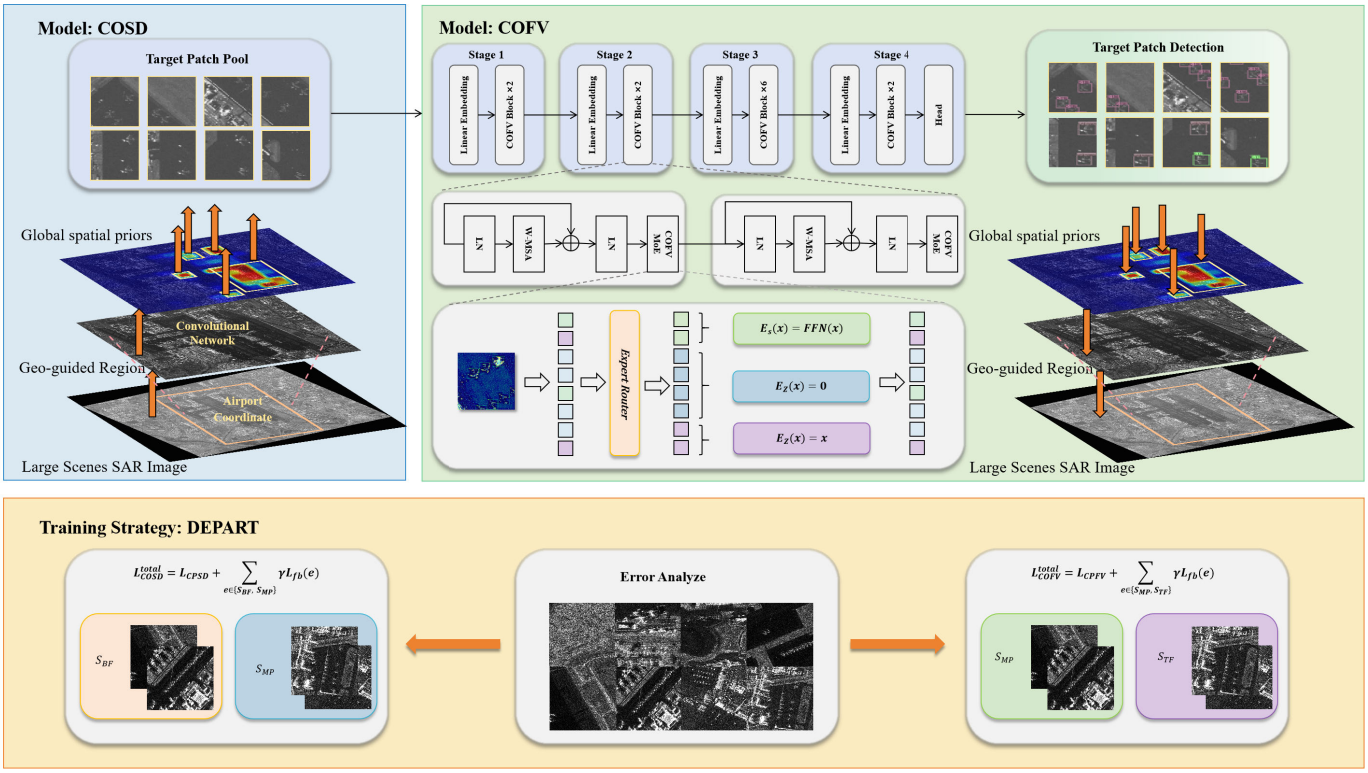


Fig. 3. Overview of CORE-Det, a framework achieving multilevel sparse perception through COSD for spatial-level patch extraction, COFV for feature-level sparsity via MoE routing, and DEPART for sample-level optimization using error-driven feedback.

extensive homogeneous regions dominated by speckle noise. How to leverage the dynamic routing mechanism of mixture-of-experts models to direct high-value scattering responses toward discriminative experts while assigning clutter and speckle-dominant features to low-cost identity experts, thereby realizing on-demand feature-level computational reallocation, remains an open problem in large-scene SAR detection. These observations motivate a unified design that connects scene-level pruning, SAR-aware feature routing, and targeted hard-sample feedback within a single framework.

III. Method

A. Overview of CORE-Det

CORE-Det addresses the limitations of conventional detectors, which lack global awareness of large-scene sparsity and tend to allocate computation almost uniformly across the scene. In SAR imagery, targets are sparse, whereas complex electromagnetic scattering, coherent speckle noise, and multipath responses frequently generate target-like clutter. CORE-Det therefore uses large-scene sparsity awareness as its starting point and information-computation alignment as its design principle, dynamically reallocating computation across the spatial, feature, and sample domains through three coordinated modules. The overall architecture is illustrated in Fig. 3. Specifically, COSD reconstructs a dynamic global spatial prior from sparse full-scene observations and performs patch-level screening before local detection, allowing the detector to reject clutter-prone background regions before they trigger local false alarms. Its joint confidence mechanism combines the global spatial prior confidence with the local detection confidence to enforce collaborative decision-making between scene-level constraints and local semantic evidence. COFV then performs SAR-tailored feature routing, sending target-dominant responses caused by strong coherent scattering and dihedral reflections to specialist experts, preserving medium-value structural context in identity experts, and routing homogeneous speckle-dominant or clutter-only responses to low-cost zero experts. DEPART adaptively selects informative hard cases through error decoupling and feeds the resulting supervision back to COSD and COFV, forming a closed-loop optimization process that sharpens their discrimination of SAR scattering patterns.

B. Computation Allocation via Spatial Distribution

Unlike pipelines that depend on external static geographic priors, COSD constructs a dynamic global spatial prior directly from the current SAR image. This design is motivated by a key property of large-scene SAR imagery: although true targets are sparse, background regions with strong scattering, multipath responses, or structural corners can still generate target-like clutter and trigger false alarms if they are processed uniformly. COSD therefore deploys sparse observation points across the entire image to estimate information density with local perception windows and reconstructs these discrete responses into a continuous global spatial prior through Gaussian diffusion fusion. This dynamically generated prior not only enables full-scene sparsity awareness and patch-level screening of low-information-density regions, but also suppresses

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TABLE I
COMPARISON OF REPRESENTATIVE SAR DETECTION DATASETS

Dataset	Images	Mean Size	Classes
MSAR [32]	28,449	256–2048 px	4
SAR-Aircraft-1.0 [33]	4,368	800–1500 px	7
SARDet-100K [34]	116,598	512×512 px	6
FAIR-CSAR [36]	14,665	1024×1024 px	5 / 22
AIR-SARShip-1.0 [35]	31	$\sim 3000 \times 3000$ px	1
FUSAR-Airport (ours)	135	$8{,}253 \times 6{,}006$ px	5
FUSAR-Harbor (ours)	60	$14{,}897 \times 11{,}646$ px	1

clutter-prone background areas before local detection and works with the local detector through a joint confidence mechanism to determine the final target confidence.

1) Global Spatial Prior Estimation: To perceive the spatial distribution of the full scene, COSD deploys N_{obs} observation points on a regular grid over the input image $\mathbf{I}_{raw} \in \mathbb{R}^{H_{total} \times W_{total} \times C}$. Each observation point n, centered at $\mathbf{c}_n = (i_n s_w,\ j_n s_h)$, extracts a local patch within a perceptual window of size $h_w \times w_w$ and estimates its local information density

$$\hat{P}_n = \Phi\left(\mathbf{I}_{raw} \mid \mathbf{c}_n,\ h_w,\ w_w,\ \mathbf{W}_\Phi\right) \tag{1}$$

where (s_w, s_h) denote the horizontal and vertical observation point spacing, $i_n, j_n \in \mathbb{Z}_{\geq 0}$ are grid indices, and $\Phi(\cdot)$ is a lightweight convolutional neural network with pretrained weights $\mathbf{W}_\Phi$ that jointly performs local patch extraction and target existence probability estimation within the perceptual window centered at $\mathbf{c}_n$. The output $\hat{P}_n \in [0, 1]$ represents the local information density estimate at observation point n. This yields the full-scene observation set $\mathcal{P} = \{\hat{P}_n, \mathbf{c}_n\}_{n=1}^{N_{\text{obs}}}$. Specifically, $\Phi(\cdot)$ is trained as a binary patch-level classifier that distinguishes target-containing patches from false-alarm background patches. Since no-target background patches greatly outnumber target-containing ones, random sampling would yield mostly easy negatives. To obtain high-value negatives instead, we construct the training samples from the inference results of COFV on the training set. Patches that contain targets are taken as positive samples, while pure background patches mistakenly predicted as targets by COFV are taken as negative samples. The pretrained weights $\mathbf{W}_\Phi$ are then obtained by optimizing a binary cross-entropy loss on these samples.

Since the observations are spatially discrete, a Gaussian diffusion fusion strategy is employed to reconstruct a continuous global spatial prior $\mathbf{M}$ from the sparse local estimates. Each observation point serves as a diffusion source, propagating its density estimate to surrounding pixels with Gaussian decay. Contributions from all sources are aggregated via weighted superposition to form a smooth global spatial prior. Let $\bar{\mathbf{c}}_n = \mathbf{c}_n + 1/2(w_w, h_w)^\top$ denote the geometric center of observation point n, and $\Omega_{(x,y)}$ the index set of all observation points covering pixel (x, y). The pixel-level global spatial prior $\mathbf{M}(x, y)$ is defined as

$$\mathbf{M}(x, y) = \frac{\sum_{n \in \Omega_{(x,y)}} \hat{P}_n \cdot w_n(x, y)}{\sum_{n \in \Omega_{(x,y)}} w_n(x, y) + \epsilon} \tag{2}$$

where ϵ is a small positive constant for numerical stability, and $w_n(x, y)$ is the Gaussian diffusion weight from observation point n to pixel (x, y)

$$w_n(x, y) = \exp\left(-\frac{\left(x - \bar{c}_{n,x}\right)^2 + \left(y - \bar{c}_{n,y}\right)^2}{2\lambda^2}\right) \tag{3}$$

where λ is a smoothing coefficient controlling the diffusion radius. This design assigns maximal contribution near the observation center, with influence decaying smoothly with distance, effectively suppressing boundary discontinuities and yielding the continuous global spatial prior $\mathbf{M}$.

2) Spatial Contextual Constraint and Patch Selection: The global spatial prior $\mathbf{M}$ encodes full-scene target-location information, compensating for the limited perception of local observations. COSD first performs coarse screening on the observation set $\mathcal{P}$ with a fixed threshold T, retaining high-information-density candidate patches satisfying $\hat{P}_n \geq T$ and forwarding them to COFV to obtain the local target detection confidence $\hat{D}_n$. The local detection confidence is then multiplied by the corresponding global spatial prior to yield the joint confidence

$$\tilde{P}_n = \hat{D}_n \cdot \mathbf{M}\left(\bar{c}_{n,x},\ \bar{c}_{n,y}\right). \tag{4}$$

The joint confidence $\tilde{P}_n$ integrates local appearance discrimination with the global spatial prior, enabling synergistic decision-making between local and global perception, and ultimately producing accurate target detection results. The product-based formulation can be interpreted as the joint probability of two co-occurring events, namely, that the local patch contains a target and that its spatial position is a plausible target region under the global spatial prior. A high joint confidence is therefore granted only when both sources of evidence are consistent, which is in line with the requirement of large-scene SAR detection that a reliable detection must be supported by both local and global evidence. Experimental results reported later show that when COSD is applied alone, the recall of all baselines improves consistently, e.g., from 76.21% to 86.90% for Oriented RCNN and from 84.07% to 89.72% for Rotated RetinaNet. This is because the spatial prior $\mathbf{M}$ is trained to suppress regions most likely to cause false alarms rather than to highlight regions of high target confidence, so that the multiplication mainly attenuates background-dominated patches while preserving genuine target patches, and the relative evidence of true targets is amplified rather than weakened.

C. Computation Allocation via Feature Value

COFV is designed to eliminate residual intrapatch computational redundancy, achieving fine-grained, feature-level resource allocation driven by information value. This design is motivated by the SAR imaging mechanism rather than by generic visual sparsity alone. In SAR scenes, informative target responses are often concentrated in a few dominant scattering centers generated by strong coherent scattering and dihedral reflections from metallic structures, whereas large homogeneous areas are dominated by coherent speckle noise and weak background returns. To exploit this property, COFV

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replaces traditional feed-forward networks (FFNs) in the backbone with MoE-based adaptive layers, enabling the network to route computation according to SAR-specific scattering characteristics while reducing invalid computational overhead.

1) Information-Value-Driven Expert Design: For the input feature sequence $\mathbf{X} = [x_1, \ldots, x_L]^\top \in \mathbb{R}^{L\times C}$, COFV constructs an MoE-based dynamic routing mechanism over a heterogeneous expert set $\mathcal{E} = \{E_S, E_I, E_Z\}$ of size $|\mathcal{E}|$, with expert parameters $\{w_0, \ldots, w_{|\mathcal{E}|-1}\}$. For each feature x_i, the router first generates a logits vector via linear projection $\mathbf{z}_i = \mathbf{W}_r x_i$, which is then dispatched to the experts through a gating function and top-K selection. The MoE layer output is defined as

$$y_i = \sum_{k=0}^{|\mathcal{E}|-1} \underbrace{\text{Softmax}\left(\mathbf{z}_i\right)_k}_{\text{Gating}} \cdot \underbrace{\mathcal{I}_K\left(\mathbf{z}_i\right)_k}_{\text{Top-}K} \cdot E_k\left(x_i, w_k\right) \tag{5}$$

where $\mathcal{I}_K(\mathbf{z}_i)_k$ is the top-K indicator function, taking value 1 if $z_{i,k}$ ranks among the top-K entries of $\mathbf{z}_i$ and 0 otherwise. This mechanism ensures each feature activates only a sparse subset of experts, satisfying the sparse computation constraint. The three expert types and their corresponding information value mappings are defined as follows.

1) *Specialist Expert (E_S):* Defined as $E_S(x_i) = \text{FFN}(x_i)$. This path is dedicated to high-value SAR responses dominated by strong coherent scattering and dihedral reflections, such as the bright returns produced by metallic aircraft structures, ship bodies, or salient man-made corners. These responses contain the most discriminative target evidence and therefore require deep nonlinear transformations to enhance fine-grained representational capacity.
2) *Identity Expert (E_I):* Defined as $E_I(x_i) = x_i$. This path corresponds to medium-value structured contextual responses, such as runway or apron edges, berth boundaries, coastline contours, and weak extended scattering adjacent to true targets. These responses are not sufficiently discriminative to justify heavy nonlinear transformation, but they carry geometric and environmental cues that help distinguish true target scattering from background-generated target-like clutter. Preserving the original feature flow while bypassing expensive matrix operations maintains this contextual support at minimal computational cost.
3) *Zero Expert (E_Z):* Defined as $E_Z(x_i) = \mathbf{0}$. This path corresponds to information-sparse regions such as open sea surfaces, runways, and homogeneous terrain, where coherent speckle noise dominates and the activations carry little discriminative value. Instead of allowing these noise-dominant responses to propagate through the network, the zero expert dynamically masks them at the source, effectively truncating invalid interference features before they accumulate into false alarms.

This information-value-driven ternary expert architecture enables COFV to adaptively construct an SAR-tailored computation graph at feature granularity, applying deep representation enhancement to target-dominant scattering features, low-cost preservation to structured contextual features, and dynamic masking to speckle-dominant noise features. In this way, COFV simultaneously compresses computational redundancy and suppresses scattering-induced false activations in complex SAR scenes.

2) Routing Gradient Estimation: The inherently discrete and nondifferentiable nature of the top-K operator in conventional MoE prevents gradients from being effectively back-propagated to the router, introducing optimization bias. To enable end-to-end joint optimization of the router and experts, we model routing gradient estimation within the numerical framework of ordinary differential equations [31]. Specifically, we adopt the discrete-time viewpoint of ODEs and treat the routing probability $\mathbf{p}$ as a continuous state evolving along expert paths, so that the otherwise nondifferentiable routing gradient can be approximated by ODE numerical schemes. A first-order Euler scheme is then applied to the optimal expert path to ensure smooth convergence of the decision boundary, while a third-order Heun scheme is applied to the remaining candidate paths to better capture higher-order nonlinear responses and preserve expert assignment diversity. The mixed-scaling output o_{est} integrates these two schemes within a single forward computation. Here, E_k denotes the kth expert in the heterogeneous set $\mathcal{E} = \{E_S, E_I, E_Z\}$, and $f(\cdot)$ denotes the downstream response with respect to the activated expert output. For an arbitrary token, with a slight abuse of notation, we drop the token subscript and let $\mathbf{z} \in \mathbb{R}^{|\mathcal{E}|}$ denote its router logits and $\mathbf{p} = \text{Softmax}(\mathbf{z})$ the corresponding routing probability. The total derivative of the training objective $\mathcal{L}$ with respect to $\mathbf{z}$ is

$$\nabla_{\mathbf{z}}\mathcal{L} = \sum_{k=0}^{|\mathcal{E}|-1} \left[p_k \frac{\partial f\left(p_k E_k\right)}{\partial \mathbf{z}} + \left(f\left(p_k E_k\right) - f\left(0\right)\right) \frac{\partial p_k}{\partial \mathbf{z}} \right] \tag{6}$$

where $f(0)$ is the baseline response representing the system output when a given expert path is inactive, and the difference term $(f(p_k E_k) - f(0))$ characterizes the instantaneous energy state transition as a feature transitions from masked to activated. Two ODE numerical schemes are introduced for continuous approximation. The first-order Euler scheme offers excellent numerical stability, where $D \sim \mathbf{p}$ is the index of the expert path under consideration, sampled according to the routing probability $\mathbf{p}$

$$\nabla_{\mathbf{z}}^{1\text{st}} = \mathbb{E}_{D\sim\mathbf{p}} \left[2 \cdot \frac{\partial f\left(p_D \cdot E_D\right)}{\partial \mathbf{z}} \right]. \tag{7}$$

The third-order Heun scheme captures higher-order nonlinear responses through multipoint evaluation, introducing a Bernoulli random variable $B \sim \text{Bernoulli}(5/8)$ as an auxiliary random variable that controls the position of the additional evaluation point used by the third-order Heun scheme

$$\nabla_{\mathbf{z}}^{3\text{rd}} = \mathbb{E}_{D,B} \left[(6 - 4B) \cdot \frac{\partial f\left(\frac{1+2B}{3} p_D E_D\right)}{\partial \mathbf{z}} \right]. \tag{8}$$

To balance training stability and expert assignment diversity, COFV applies the first-order Euler approximation to the optimal expert path ($D = \arg\max(\mathbf{z})$) to ensure smooth convergence of the decision boundary, and the third-order Heun approximation to all remaining candidate paths.

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End-to-end optimization is achieved through a mixed-scaling output o_{est}

$$o_{est} = 2 \cdot o + \text{detach}\left(\max\left(\delta_D, \tfrac{1+2B}{3}\right) \cdot o - 2 \cdot o\right) \tag{9}$$

where $o = p_D \cdot E_D(x)$ is the forward output of the sampled expert after gating weight scaling, δ_D is the maximum probability indicator function, and detach$(\cdot)$ is the gradient stop operator. This mechanism enables COFV to automatically learn, under sparse activation constraints, to route low-value background features to identity experts and pure-noise features to zero experts, realizing information-value-aligned on-demand feature-level computational reallocation.

D. Decoupled Error Partitioning for Adaptive Retraining

In large-scene SAR target detection, the extreme sparsity of targets causes severe positive-negative sample imbalance. More importantly, many hard cases arise from SAR-specific scattering ambiguity: some background regions generate target-like clutter, whereas some true targets exhibit weak or partially suppressed scattering responses. Traditional training strategies oscillate between the extremes of full-scene and target-patch sampling, failing to adaptively focus on these high-value hard cases and lacking targeted optimization feedback for COSD and COFV. To address this, we propose the DEPART module, which decouples the responsibility attribution of inference failures to establish a directed feedback mapping from error types to their corresponding modules. This mechanism drives the continuous iterative optimization of COSD and COFV within a dynamically reconstructed pool of informative hard cases, enabling the two modules to better discriminate SAR scattering features and ultimately forming a closed-loop adaptive learning system.

1) Decoupled Error Attribution: DEPART dynamically constructs a high-value hard case pool using current-iteration inference failures. Following forward inference on large-scene SAR images, failed instances are decoupled into the overall error set $\mathcal{S}_{err} = \{\mathcal{S}_{BF}, \mathcal{S}_{MP}, \mathcal{S}_{TF}, \mathcal{S}_{MD}\}$, which includes background region false alarms, pruning misses, target region false alarms, and detector misses.

1) $\mathcal{S}_{BF}$*:* False alarms originating from pure background regions. In SAR imagery, these cases are often induced by strong background scatterers that generate target-like clutter. This indicates an overestimated background confidence by COSD, causing invalid patches to bypass the filter, attributing responsibility to COSD.
2) $\mathcal{S}_{MP}$*:* Target patches mistakenly discarded as background during coarse screening. In SAR imagery, this often occurs when weak target scattering is submerged by surrounding clutter or homogeneous background responses. This reveals an insufficient response from COSD to target regions, leading to the premature loss of valid targets, attributing responsibility to COSD.
3) $\mathcal{S}_{TF}$*:* False alarms induced by complex strong scatterers. This reflects COFV's lack of fine-grained discrimination in local high-response areas, where clutter-dominant scattering is not adequately separated from true target scattering, attributing responsibility to COFV and necessitating redirection to the zero expert or the identity expert according to the actual information value.
4) $\mathcal{S}_{MD}$*:* Target patches failing to be identified after entering COFV. This exposes the inadequate feature routing of COFV, where weak but informative target scattering or supportive contextual scattering is not sent to the appropriate experts, attributing responsibility to COFV and requiring more suitable routing to specialist or identity experts.

2) Module-Directed Feedback Loss: Based on this attribution, DEPART establishes a directed feedback mapping to each module. The COSD branch leverages $\mathcal{S}_{BF}$ and $\mathcal{S}_{MP}$ to suppress background clutter and elevate target saliency, widening the discriminative margin between target-bearing regions and clutter-prone backgrounds. Simultaneously, the COFV branch utilizes $\mathcal{S}_{TF}$ and $\mathcal{S}_{MD}$ to guide expert selection: activating E_Z for noise-only interference suppression, preserving structurally useful responses in E_I, and activating E_S for target-dominant feature enhancement. The overall optimization objective is composed of the base supervision loss and the corresponding feedback loss for each module

$$\mathcal{L}_{total} = \mathcal{L}_{COSD}^{total} + \mathcal{L}_{COFV}^{total} \tag{10}$$

$$\mathcal{L}_{COSD}^{total} = \mathcal{L}_{COSD} + \sum_{e \in \{\mathcal{S}_{BF}, \mathcal{S}_{MP}\}} \gamma_e \mathcal{L}_{fb}(e) \tag{11}$$

$$\mathcal{L}_{COFV}^{total} = \mathcal{L}_{COFV} + \sum_{e \in \{\mathcal{S}_{TF}, \mathcal{S}_{MD}\}} \gamma_e \mathcal{L}_{fb}(e) \tag{12}$$

where $\mathcal{L}_{COSD}$ and $\mathcal{L}_{COFV}$ denote the base supervision losses of the two modules, such as standard classification loss and bounding box regression loss, $\mathcal{L}_{fb}(e)$ represents the feedback penalty for error instance e, and γ_e is a dynamic weight coefficient. Specifically, feedback from $\mathcal{S}_{BF}$ and $\mathcal{S}_{MP}$ drives the global spatial prior $\mathbf{M}$ toward a bimodal state to enhance the robustness of threshold T and to better suppress background-generated target-like clutter. Meanwhile, feedback from $\mathcal{S}_{TF}$ and $\mathcal{S}_{MD}$ guides the router to refine the mixed-scaling output o_{est} for precise assignment among E_S, E_I, and E_Z, thereby increasing the separation between target-dominant scattering, structured contextual scattering, and noise-dominant responses. Through this modular error-attribution loss mechanism, CORE-Det adaptively converges toward the optimal decision boundary across the full scene and forms a closed-loop optimization system.

IV. Experiments

A. Datasets

To evaluate sparsity-aware detection on full-scene SAR imagery, it is important to distinguish between generic SAR detection benchmarks and datasets that preserve end-to-end large-scene sparsity. As summarized in Table I, many widely used datasets, including MSAR [32], SAR-Aircraft-1.0 [33], and SARDet-100K [34], are organized around fixed-size slices or target-centered crops. AIR-SARShip-1.0 [35] contains scene-level ship imagery, but it is single-class and relatively limited in scene scale. FAIR-CSAR [36] substantially advances fine-grained SAR object detection and recognition,

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TABLE II
PERFORMANCE COMPARISON OF BASELINE DETECTORS WITH AND WITHOUT CORE-DET ON THE FUSAR-AIRPORT

Method	CORE-Det	Overall Metrics			AP_{50}						Efficiency		
		mF1	Recall	FA	mAP_{50}	LA	OA	PA	SA	TA	Params (M)	GFLOPs	Time
Oriented RCNN		0.6613	0.7621	0.2974	0.6553	**0.8399**	0.5411	0.4823	0.6828	0.7303	44.75	26,176.95	6.96 s
	✓	**0.7896**	**0.8669**	**0.0909**	**0.7480**	0.8012	**0.7547**	**0.5955**	**0.8349**	**0.7538**	46.30	**3,540.94**	**2.99 s**
RoI-Trans		0.7334	0.7722	0.1470	0.7274	0.8324	0.6705	0.6077	0.8231	0.7035	58.68	27,850.71	6.32 s
	✓	**0.7964**	**0.8690**	**0.0610**	**0.7716**	**0.8848**	**0.7391**	**0.6410**	**0.8571**	**0.7358**	60.22	**3,900.54**	**2.68 s**
Rotated Faster RCNN		0.6997	0.7601	0.2096	0.7007	0.8257	0.6435	0.4883	0.7838	**0.7622**	44.75	26,159.95	5.69 s
	✓	**0.7778**	**0.8548**	**0.0848**	**0.7306**	**0.8523**	**0.7110**	**0.6077**	**0.8383**	0.6437	46.29	**3,628.82**	**2.89 s**
Rotated ATSS		0.5963	0.5282	0.1683	0.6350	0.7836	0.4965	**0.5624**	0.5980	0.7344	36.74	25,918.42	8.79 s
	✓	**0.7704**	**0.7944**	**0.0901**	**0.7452**	**0.9054**	**0.6858**	0.5481	**0.8207**	**0.7658**	38.28	**3,677.29**	**3.21 s**
Rotated FCOS		0.7339	0.7440	0.1800	0.7440	**0.8673**	0.6288	**0.6591**	0.7459	**0.8188**	35.56	25,892.93	5.68 s
	✓	**0.7680**	**0.7984**	**0.0979**	**0.7494**	0.8507	**0.7260**	0.6251	**0.7984**	0.7467	37.11	**3,495.75**	**2.13 s**
Rotated RetinaNet		0.6773	0.8407	0.3454	0.6350	0.7836	0.4965	0.5624	0.5980	0.7344	36.92	26,407.56	9.03 s
	✓	**0.8091**	**0.8750**	**0.0863**	**0.7944**	**0.8765**	**0.7628**	**0.6786**	**0.8551**	**0.7989**	38.47	**3,847.05**	**3.03 s**

yet it is not primarily designed as an efficiency benchmark for end-to-end sparse scheduling in very large airport and harbor scenes. Consequently, existing public datasets do not fully reflect the joint spatial, feature-level, and sample-level sparsity encountered in computation-aware large-scene SAR detection. We therefore construct two full-scene datasets, FUSAR-Airport and FUSAR-Harbor, to enable direct evaluation on large-scale imagery. Both datasets are split in a location-stratified manner: scenes from each location are partitioned between the training and validation subsets at a ratio of 8–2, ensuring that every location appears in both subsets so that the validation results are not biased by geographic distribution. The validation images are further chosen to maintain balanced coverage across the GF-3 and QL-1 satellites, the HH and VV polarizations, and the five imaging strips KRN, MYN, KAS, MDJ, and SAY. The validation set therefore jointly reflects the diversity of acquisition platforms, polarization conditions, and scene types in the data, providing a fair and representative basis for performance evaluation.

1) FUSAR-Airport: This dataset comprises 135 full-scene images, including 113 sourced from GF-3 and 22 from QL-1. The total number of annotated target instances is 3608, yielding an average of approximately 26.73 instances per image with a mean resolution of 8253 × 6006 pixels (width × height). The targets are categorized into five classes: 1143 other aircraft (OA), 775 small aircraft (SA), 745 large aircraft (LA), 557 transport aircraft (TA), and 388 special-purpose aircraft (PA). Featuring complex terrestrial backgrounds and densely arranged aircraft, this dataset serves as the primary benchmark for ablation studies and performance analysis in this work.

2) FUSAR-Harbor: This dataset consists of 60 full-scene images, totaling 45 from GF-3 and 15 from QL-1, with a mean image size of 14 897 × 11 646 pixels. A single-class annotation strategy is adopted for ships, encompassing a total of 1004 instances. Designed for ship detection in complex coastal environments with intricate sea-land boundaries, this dataset is primarily employed to evaluate the cross-scenario generalization capability of the proposed model.

B. Baseline Methods and Implementation Details

To validate the universal adaptability of the CORE-Det framework across diverse detection architectures, six representative oriented object detection methods are selected as baselines, spanning both two-stage and single-stage paradigms. The two-stage detectors comprise Oriented RCNN [21], RoI Transformer (RoI-Trans) [20], and Rotated Faster RCNN; the single-stage detectors include Rotated ATSS, Rotated FCOS, and Rotated RetinaNet [17]. To ensure a fair comparison, all baseline methods uniformly adopt Swin-Tiny [19] as the backbone network. All experiments are implemented within the MMRotate framework, with both training and inference conducted on a single NVIDIA A800 GPU. For all baseline detectors, both two-stage and one-stage, the COSD screening threshold is uniformly set to $T = 0.5$, and the sensitivity of this choice is further examined in Section IV-D1. For the top-K routing in COFV, we adopt $K = 1$ throughout all experiments, so that each feature token activates exactly one expert from $\{E_S, E_I, E_Z\}$. To further control the comparison protocol, all baseline detectors and their CORE-Det-enhanced counterparts share the same geo-guided region extraction before local detection. Therefore, the geographic prior used to reduce the search space is not an exclusive advantage of CORE-Det in the comparative experiments; instead, the reported gains reflect the additional computation reallocation introduced after this common preprocessing stage.

The following metrics are employed to evaluate detection performance. Recall, FA, macro-averaged $F1$ score (mF1), and mean average precision (mAP) are, respectively, defined as

$$\text{Recall} = \frac{N_{TP}}{N_{TP} + N_{FN}} \tag{13}$$

$$\text{FA} = \frac{N_{FP}}{N_{FP} + N_{TP}} \tag{14}$$

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TABLE III
CROSS-SCENARIO GENERALIZATION EXPERIMENTAL RESULTS ON THE FUSAR-HARBOR

Method	CORE-Det	Overall Metrics			AP_{50}	Efficiency		
		mF1	Recall	FA	mAP_{50}	Params (M)	GFLOPs	Time
Oriented RCNN		0.7210	0.7823	0.3314	0.7894	44.75	48,679.80	7.10 s
	✓	**0.8521**	**0.8231**	**0.1168**	**0.8256**	46.30	**4,124.84**	**5.23 s**
RoI-Trans		0.7266	0.6599	0.1917	0.7582	58.68	51,792.40	7.47 s
	✓	**0.8222**	**0.7551**	**0.0976**	**0.7810**	60.22	**3,601.57**	**4.34 s**
Rotated Faster RCNN		0.5579	0.6395	0.5053	0.5012	44.75	48,648.20	7.57 s
	✓	**0.7660**	**0.7347**	**0.2000**	**0.6668**	46.29	**4,703.33**	**5.40 s**
Rotated ATSS		0.4961	0.4422	0.4348	0.4365	36.74	48,199.03	22.23 s
	✓	**0.6246**	**0.5374**	**0.2547**	**0.5618**	38.28	**3,944.68**	**4.62 s**
Rotated FCOS		0.6407	0.7823	0.4575	0.7334	35.56	48,151.63	6.57 s
	✓	**0.8163**	**0.8163**	**0.1837**	**0.7874**	37.11	**3,416.53**	**4.87 s**
Rotated RetinaNet		0.4203	0.5918	0.6742	0.3520	36.92	49,108.66	13.53 s
	✓	**0.7085**	**0.6531**	**0.2258**	**0.6568**	38.47	**4,392.77**	**4.69 s**

$$\mathrm{mF1} = \frac{1}{C}\sum_{c=1}^{C}\frac{2\cdot \mathrm{Precision}_c \cdot \mathrm{Recall}_c}{\mathrm{Precision}_c + \mathrm{Recall}_c} \tag{15}$$

$$\mathrm{mAP} = \frac{1}{C}\sum_{c=1}^{C}\int_0^1 P_c(R)\, dR \tag{16}$$

where N_{TP}, N_{FP}, and N_{FN} denote the number of true positives, false positives, and false negatives, respectively; C is the total number of categories; and $P_c(R)$ represents the precision-recall curve for category c. Recall quantifies the detector's coverage of ground-truth targets, while FA measures the proportion of spurious detections among all positive predictions; these two metrics characterize the detector's performance boundaries from the complementary perspectives of missed detections and false alarms. mF1 computes the harmonic mean of precision and recall at a given confidence threshold and macro-averages over all categories, thereby reflecting the system's overall detection performance at a specific operating point. In contrast, mAP integrates the precision-recall curve across the full threshold range, providing a threshold-agnostic global performance characterization. In addition, full-scene FLOPs and model parameter count, reported as Params (M), are used to assess computational efficiency. For Tables II and III, the AP-based results are reported at an IoU threshold of 0.5 and are therefore denoted as AP_{50} and mAP_{50}.

C. Overall Performance Evaluation

This section evaluates CORE-Det on the FUSAR-Airport and FUSAR-Harbor datasets in terms of detection accuracy, computational efficiency, and cross-scene behavior. FUSAR-Airport contains five aircraft categories in airport scenes, whereas FUSAR-Harbor focuses on single-class ship detection in complex coastal environments. Unless otherwise stated, the Time metric denotes the sum of model inference and nonmaximum suppression on a single NVIDIA A800 GPU. Table II reports the comparison between six baseline detectors and their CORE-Det-enhanced counterparts on FUSAR-Airport. The results are obtained on a fixed 16-image validation split defined at the full-scene image level. All methods use the same split, and each validation image covers a large spatial extent with many annotated targets and diverse background structures, which provides a stable basis for model comparison. Because the baseline detectors already operate with the same geo-guided patch extraction protocol, the improvements observed in Tables II and III should be attributed to the dynamic spatial prior reconstruction, SAR-aware feature routing, and error-driven retraining of CORE-Det rather than to unequal access to geographic priors.

1) Performance Evaluation on FUSAR-Airport: On FUSAR-Airport, CORE-Det consistently improves the accuracy-efficiency tradeoff across all six baselines. For example, when integrated into Oriented RCNN, CORE-Det increases recall from 76.21% to 86.69% and reduces the false alarm rate from 29.74% to 9.09%. Similar behavior is observed for the single-stage Rotated RetinaNet, whose recall increases from 84.07% to 87.50% while the false alarm rate decreases from 34.54% to 8.63%. The efficiency gains are also substantial. For Oriented RCNN, GFLOPs decrease from 26 176.95 to 3540.94, and the end-to-end processing time decreases from 6.96 to 2.99 s. Averaged over the six baselines, CORE-Det reduces GFLOPs by 86.06% and end-to-end processing time by 59.38%, while improving mAP_{50}, mF1, and recall by 7.36, 10.16, and 10.85 percentage points (pp), respectively, and reducing false alarms by 13.95 pp. These results indicate that CORE-Det improves both detection quality and computational efficiency on large airport scenes under the evaluated setting.

2) Performance Evaluation on FUSAR-Harbor: To further assess cross-scene behavior, we evaluate CORE-Det on the FUSAR-Harbor dataset. Compared with FUSAR-Airport, FUSAR-Harbor contains different targets, more complex sea-land interfaces, and denser coastal clutter. Table III summarizes the corresponding comparison results.

On FUSAR-Harbor, CORE-Det again yields consistent gains across all six baselines. For example, when integrated into Oriented RCNN, recall increases from 78.23% to 82.31%,

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TABLE IV
COSD EFFECTIVENESS EXPERIMENT ON THE FUSAR-AIRPORT

Method	COSD	Overall Metrics			Efficiency	
		mF1	Recall	FA	GFLOPs	Params (M)
Oriented RCNN		0.6613	0.7621	0.2974	26,176.95	44.75
	✓	**0.7347**	**0.8690**	**0.1631**	**4,565.66**	46.28
RoI-Trans		0.7334	0.7722	0.1470	27,850.71	58.68
	✓	**0.7715**	**0.8770**	**0.0919**	**4,831.98**	60.21
Rotated Faster RCNN		0.6997	0.7601	0.2096	26,159.95	44.75
	✓	**0.7564**	**0.8589**	**0.1377**	**4,562.96**	46.28
Rotated ATSS		0.5963	0.5282	0.1683	25,918.42	36.74
	✓	**0.7174**	**0.7379**	**0.1408**	**4,524.53**	38.27
Rotated FCOS		0.7339	0.7440	0.1800	25,892.93	35.56
	✓	**0.7886**	**0.7903**	**0.1151**	**4,520.47**	37.09
Rotated RetinaNet		0.6773	0.8407	0.3454	26,407.56	36.92
	✓	**0.7301**	**0.8972**	**0.2657**	**4,602.36**	38.45

the false alarm rate decreases from 33.14% to 11.68%, and mF1 reaches 85.21%. Likewise, for Rotated RetinaNet, the false alarm rate decreases from 67.42% to 22.58%, indicating that the spatial prescreening and feature routing mechanisms remain effective in cluttered harbor scenes. The efficiency gains are even more pronounced on FUSAR-Harbor because of the larger image sizes and sparser target distribution. For RoI-Trans, GFLOPs decrease from 51 792.40 to 3601.57, corresponding to a 93.0% reduction. The reductions in redundant proposals also alleviate the NMS bottleneck of one-stage detectors: the end-to-end latency of Rotated ATSS and Rotated RetinaNet drops from 22.23 to 4.62 s and from 13.53 to 4.69 s, respectively. Averaged over the six baselines, CORE-Det reduces GFLOPs by 91.78% and end-to-end processing time by 44.56%, while improving mAP_{50}, mF1, and recall by 11.81, 17.12, and 7.03 pp, respectively, and reducing false alarms by 25.27 pp. Taken together, these results provide evidence that CORE-Det remains effective across the evaluated airport and harbor scenarios.

D. Ablation Studies

To validate the independent effectiveness and synergistic gains of each module in CORE-Det, this section conducts ablation studies on FUSAR-Airport. For large-scene SAR detection, recall and false alarm rate are the two most deployment-relevant indicators, while mF1 provides a unified measure of their balance. We therefore use mF1 as the primary criterion and report recall and false alarm rate alongside it to show how each module improves target coverage and clutter suppression. The experimental validations detailed in this section systematically encompass the effectiveness of COSD, COFV, and DEPART, and a hierarchical ablation on the CORE-Det.

1) Effectiveness of COSD: To evaluate how COSD addresses spatial sparsity, we study the module from both quantitative and qualitative perspectives. We first insert COSD into six baseline detectors and compare the resulting full-scene efficiency and detection performance. We then visualize the dynamic global spatial priors generated by COSD to examine how the module responds to scene structure and temporal variation. Finally, we conduct a quantitative sensitivity analysis on the screening threshold T to justify the value used in the main experiments.

As shown in Table IV, COSD reduces the end-to-end computational burden of both single- and two-stage detectors from about 2.6×10^4 to 4.5×10^3 GFLOPs, corresponding to an average 5.7× reduction. At the same time, detection quality improves. For Rotated ATSS, recall increases from 52.82% to 73.79%, together with a 12.11 pp gain in mF1. For the two-stage Oriented RCNN, recall rises from 76.21% to 86.90% while the false alarm rate decreases from 29.74% to 16.31%. These results indicate that COSD removes large amounts of low-value background computation while suppressing background-generated target-like clutter and preserving the scene regions that are most likely to contain true targets. Fig. 4 visualizes the dynamic global spatial prior generated by COSD. High responses concentrate around aircraft stands, terminal-side apron regions, and other locations where target-related scattering is repeatedly observed, whereas open pavement and empty parking positions are suppressed even when they share similar coarse geometry. This behavior suggests that COSD does not rely on a static airport template; instead, it reconstructs a content-dependent prior from the current SAR observation by combining scene structure with local target evidence, so that clutter-prone background regions can be downweighted before local detection. Across multitemporal views of the same airport, the high-response regions shift with aircraft relocation, showing that the prior follows scene-dependent scattering changes rather than fixed coordinates. As a result, COSD can perform patch-level screening before local detection while remaining adaptive to both spatial layout and temporal variation. These observations also clarify the practical sensitivity of COSD. The screening threshold T critically determines the effective retention of the reconstructed prior. Specifically, an excessively high threshold increases

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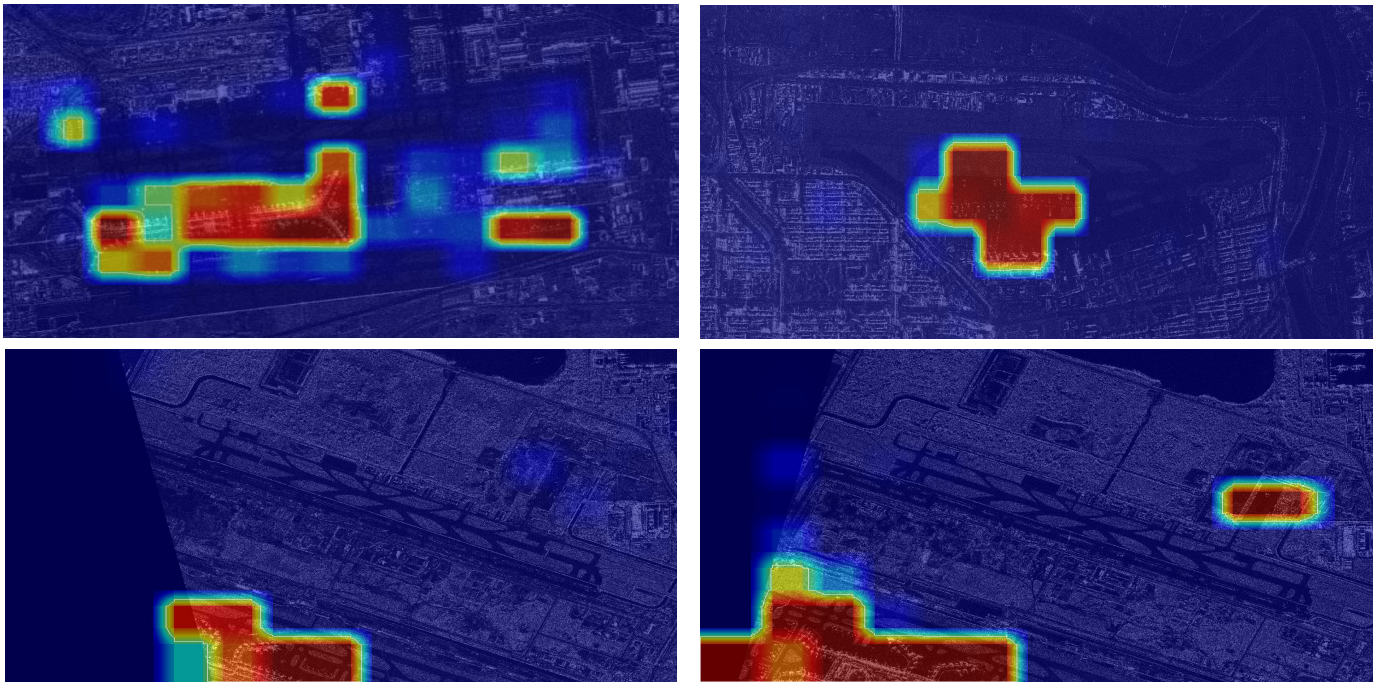


Fig. 4. Visualization of the dynamic global spatial prior **M** generated by COSD. Warmer red and cooler blue shades indicate regions with high and low target likelihood, respectively.

TABLE V

SENSITIVITY ANALYSIS OF THE COSD SCREENING THRESHOLD T WITH ORIENTED RCNN ON FUSAR-HARBOR

Threshold T	Recall (%)	FA (%)	Patches Filtered (%)	Avg Comp. (GFLOPs/img)
0.25	**87.07**	42.08	90.4	4,519.6
0.50	**87.07**	39.91	91.4	4,124.8
0.75	85.03	**38.73**	**92.3**	**3,754.8**

the risk of erroneously pruning weak targets, whereas an overly low threshold retains excessive background, thereby compromising the computational benefits of the prescreening stage. To examine this sensitivity quantitatively and to justify the value used in the main experiments, Oriented RCNN is taken as the representative detector, and three thresholds, $T \in \{0.25, 0.5, 0.75\}$, are evaluated on FUSAR-Harbor under an identical protocol, with the NMS IoU set to 0.1 and the score threshold set to 0.05. Table V reports the recall, false alarm rate, the proportion of filtered patches, and the average computational cost per image for each threshold. A low threshold of $T = 0.25$ retains the most patches and yields the highest computational cost and false alarm rate, without improving the recall over $T = 0.5$. A high threshold of $T = 0.75$ minimizes computation but degrades the recall from 87.07% to 85.03%, because an excessively aggressive threshold begins to discard patches that contain true targets. In comparison, $T = 0.5$ attains the same recall as $T = 0.25$ while reducing both the false alarm rate and the computational cost, and it avoids the recall loss observed at $T = 0.75$. Accordingly, $T = 0.5$ is selected as the default screening threshold for both two-stage and one-stage detectors in all main experiments. The synergistic effects of COSD, COFV, and DEPART are summarized in Table VI.

Overall, COSD provides the largest efficiency gain among the three modules. Averaged over the six baselines, it reduces GFLOPs by 82.6% and improves mF1 by 6.61 pp, driven by a 10.38 pp gain in recall and a 7.22 pp reduction in false alarms. The key benefit of COSD is the introduction of a dynamic global spatial prior that separates region-level screening from local refined detection, allowing computation to concentrate on high-value scene regions while clutter-prone background areas are pruned early.

2) Effectiveness of COFV: COFV addresses feature-level redundancy within the three-level sparsity decoupling mechanism of CORE-Det. To evaluate its routing behavior, we insert COFV independently into each baseline detector. The quantitative results are summarized in Table VII.

Using COFV alone consistently reduces false alarms and feature computation. For the two-stage Oriented RCNN, GFLOPs decrease from 26 176.95 to 20 372.79, while the false alarm rate drops from 29.74% to 25.80%. Because the routing policy is intentionally conservative during early feature extraction, some weak target responses are occasionally assigned to low-cost experts, which causes recall to decrease slightly from 76.21% to 74.80%. Even so, the reduction in clutter-induced false alarms improves mF1 from 66.13% to 67.83%. A similar trend appears for Rotated RetinaNet, where GFLOPs decrease from 26 407.56 to 21 726.12 and mF1 still improves from 67.73% to 68.37% despite a minor recall drop from 84.07% to 82.46%. Fig. 5 visualizes the spatial activation of specialist expert E_S, which is also the only expert that incurs the full FFN cost. In both airport and harbor scenes, strong activation appears around bright scattering centers, elongated ship or aircraft bodies, and man-made corner structures. These regions are consistent with SAR responses dominated by strong coherent scattering and dihedral reflections, which are exactly the patterns that COFV is designed to route to the specialist expert. By contrast, large homogeneous regions remain weakly activated in E_S, indicating that they are mostly handled by the

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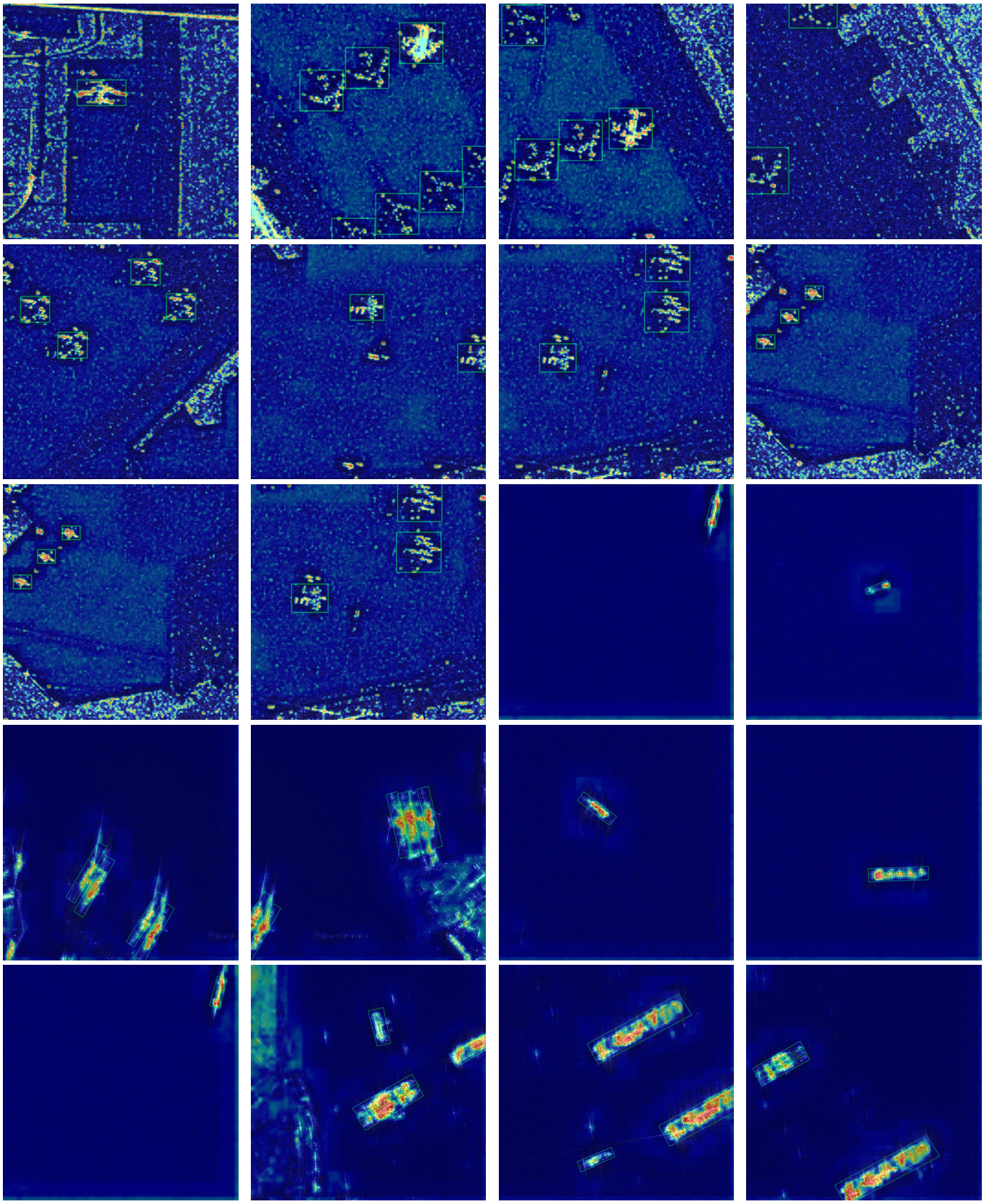


Fig. 5. Visualization of specialist expert activation E_S in COFV. Hot regions indicate where high-capacity computation is assigned to target-dominant scattering responses, whereas cool regions correspond to low-cost handling of homogeneous background features.

low-cost experts. Importantly, these low-E_S regions are not all discarded: structured contextual scattering, such as coastlines, berth edges, runway boundaries, and target-adjacent weak responses, can still be preserved through the identity expert E_I,

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TABLE VI
SYNERGISTIC ABLATION ANALYSIS

Method	COSD	DEPART	COFV	Overall Metrics			Efficiency	
				mF1	Recall	FA	GFLOPs	Params (M)
Oriented RCNN				0.6613	0.7621	0.2974	26,176.95	44.75
			✓	0.6783	0.7480	0.2580	20,372.79	44.77
		✓	✓	0.7232	0.8145	0.2140	19,736.79	44.77
	✓	✓	✓	**0.7896**	**0.8669**	**0.0909**	**3,540.94**	46.30
RoI-Trans				0.7334	0.7722	0.1470	27,850.71	58.68
			✓	0.7265	0.7440	0.1399	21,996.79	58.69
		✓	✓	0.7668	0.7964	0.1300	22,583.03	58.69
	✓	✓	✓	**0.7964**	**0.8690**	**0.0610**	**3,900.54**	60.22
Rotated Faster RCNN				0.6997	0.7601	0.2096	26,159.95	44.75
			✓	0.7112	0.7319	0.1879	20,306.04	44.76
		✓	✓	0.7283	0.7923	0.1963	20,289.04	44.76
	✓	✓	✓	**0.7778**	**0.8548**	**0.0848**	**3,628.82**	46.29
Rotated ATSS				0.5963	0.5282	0.1683	25,918.42	36.74
			✓	0.6289	0.5565	0.1210	21,238.20	36.75
		✓	✓	0.6711	0.6431	0.1493	20,593.70	36.75
	✓	✓	✓	**0.7704**	**0.7944**	**0.0901**	**3,677.29**	38.28
Rotated FCOS				0.7339	0.7440	0.1800	25,892.93	35.56
			✓	0.7156	0.6815	0.1592	20,660.45	35.58
		✓	✓	0.7203	0.6835	0.1263	19,452.77	35.58
	✓	✓	✓	**0.7680**	**0.7984**	**0.0979**	**3,495.75**	37.11
Rotated RetinaNet				0.6773	0.8407	0.3454	26,407.56	36.92
			✓	0.6837	0.8246	0.3393	21,726.12	36.94
		✓	✓	0.7160	0.8427	0.2927	21,660.58	36.94
	✓	✓	✓	**0.8091**	**0.8750**	**0.0863**	**3,847.05**	38.47

whereas speckle-dominant or purely misleading responses are routed to E_Z. The visualization therefore matches the intended SAR-aware routing behavior: high-value scattering responses receive specialized processing, contextual scattering is retained at low cost, and noise-dominant activations are suppressed before they accumulate into false alarms. From the standpoint of parameter sensitivity, the top-K routing sparsity governs a tradeoff between expert capacity and computational savings: more aggressive sparse routing removes more redundant computation, but it can also suppress weak target responses when informative contextual features are prematurely directed away from E_S. In our experiments, $K = 1$ is adopted as the most sparse setting to maximize the computational saving of COFV, and the potential suppression of weak target responses is subsequently compensated by DEPART's feedback-driven retraining, which redirects misrouted hard cases back to the appropriate experts.

In summary, the isolated application of COFV reduces the average false alarm rate by 2.37 pp and prunes 20.3% of redundant inference operations, improving the mF1 score by 0.71 pp despite a 2.01 pp drop in recall. The observed recall deficit stems from the feature router misallocating high-value weak targets to lightweight experts due to inadequate clutter suppression. DEPART mitigates this by employing feedback-driven optimization for the routing strategy. This enables the precise redirection of weak targets to specialist experts for fine-grained extraction, thereby not only recovering the lost recall but also achieving performance metrics that surpass the initial baseline.

3) Effectiveness of DEPART: To evaluate the effectiveness of the DEPART module during the training process, we conduct independent ablation studies. In SAR imagery, detection bottlenecks primarily stem from scattering-ambiguous hard cases, which serve as high-value samples for model optimization. These include background-generated target-like clutter and weak targets immersed in complex environments. The quantitative results are summarized in Table VIII.

Integrating DEPART into the baseline detectors consistently improves recall and false alarm suppression without introducing any additional inference overhead. For instance, in Oriented RCNN, DEPART increases recall from 76.21% to 82.46% and reduces the false alarm rate from 29.74% to 22.83%. This elevates the mF1 score from 66.13% to 70.30%, while the computational cost remains unchanged at 26,176.95 GFLOPs. Similarly, when applied to Rotated ATSS, recall increases from 52.82% to 61.69%, and the false alarm rate decreases from 16.83% to 14.29%, lifting mF1 from 59.63% to 65.90%.

Overall, DEPART achieves consistent performance gains across all evaluated networks. It improves recall by an average of 3.39 pp and reduces the false alarm rate by 3.24 pp, thereby increasing mF1 by 3.03 pp without affecting inference costs. This performance improvement is principally attributable to the module's capacity to compel the network

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TABLE VII
EFFECTIVENESS OF COFV ON THE FUSAR-AIRPORT

Method	COFV	Overall Metrics			Efficiency	
		mF1	Recall	FA	GFLOPs	Params (M)
Oriented RCNN		0.6613	**0.7621**	0.2974	26,176.95	44.75
	✓	**0.6783**	0.7480	**0.2580**	**20,372.79**	44.77
RoI-Trans		**0.7334**	**0.7722**	0.1470	27,850.71	58.68
	✓	0.7265	0.7440	**0.1399**	**21,996.79**	58.69
Rotated Faster RCNN		0.6997	**0.7601**	0.2096	26,159.95	44.75
	✓	**0.7112**	0.7319	**0.1879**	**20,306.04**	44.76
Rotated ATSS		0.5963	0.5282	0.1683	25,918.42	36.74
	✓	**0.6289**	**0.5565**	**0.1210**	**21,238.20**	36.75
Rotated FCOS		**0.7339**	**0.7440**	0.1800	25,892.93	35.56
	✓	0.7156	0.6815	**0.1592**	**20,660.45**	35.58
Rotated RetinaNet		0.6773	**0.8407**	0.3454	26,407.56	36.92
	✓	**0.6837**	0.8246	**0.3393**	**21,726.12**	36.94

TABLE VIII
EFFECTIVENESS OF DEPART ON THE FUSAR-AIRPORT

Method	DEPART	Overall Metrics		
		mF1	Recall	FA
Oriented RCNN		0.6613	0.7621	0.2974
	✓	**0.7030**	**0.8246**	**0.2283**
RoI-Trans		0.7334	0.7722	0.1470
	✓	**0.7610**	**0.7863**	**0.1295**
Rotated Faster RCNN		0.6997	0.7601	0.2096
	✓	**0.7151**	**0.7782**	**0.1874**
Rotated ATSS		0.5963	0.5282	0.1683
	✓	**0.6590**	**0.6169**	**0.1429**
Rotated FCOS		0.7339	0.7440	0.1800
	✓	**0.7428**	**0.7520**	**0.1599**
Rotated RetinaNet		0.6773	0.8407	0.3454
	✓	**0.7028**	**0.8528**	**0.3054**

to extract more discriminative representations from high-value scattering samples. By leveraging error-driven feedback, DEPART dynamically focuses the training phase on these scattering-ambiguous hard cases, effectively optimizing the discriminative power of the network's feature representations.

4) Synergistic Ablation Analysis: To analyze the interaction among the sparsity-aware modules in CORE-Det, Table VI reports the progressive ablation results obtained by introducing COFV, then DEPART, and finally COSD into each baseline.

This progression reveals a clear division of labor among the three modules. COFV first reduces feature-level redundancy and false alarms, but it also lowers recall relative to the baseline because conservative early stage routing can suppress weak target responses. DEPART then compensates for this effect by feeding scattering-ambiguous hard cases back to the router. For example, the recall of Oriented RCNN increases from 74.80% to 81.45%, exceeding its original baseline of 76.21%. At the same time, the GFLOPs of Rotated FCOS further decrease from 20 660.45 to 19 452.77. This behavior indicates that DEPART helps COFV recover weak but informative targets, preserve medium-value contextual scattering in E_1, and continue to reject invalid background activations. Adding COSD on top of COFV and DEPART yields the best overall operating point. For Rotated RetinaNet, mF1 increases from 71.60% to 80.91%, recall reaches 87.50%, and the false alarm rate decreases from 29.27% to 8.63%. The additional gain stems from scene-level screening and joint confidence scoring, whereby COSD removes clutter-prone low-value patches prior to local inference and constrains final decisions through the dynamic global spatial prior.

Overall, the synergistic ablation results are logically consistent with the design of CORE-Det. COFV addresses feature-level redundancy, DEPART corrects routing errors exposed by SAR scattering ambiguity, and COSD further suppresses scene-level clutter before local detection. Their combined results support the design of CORE-Det as a coordinated spatial-feature-sample reallocation framework for large-scene SAR detection. The consistent gains across six baselines and two scene types also suggest that CORE-Det is not tied to a single backbone or operating point, although the absolute gain still depends on target sparsity and scene complexity.

V. CONCLUSION

We presented CORE-Det, a computation reallocation framework for large-scene SAR target detection. CORE-Det addresses this mismatch through three complementary components, wherein COSD reconstructs a dynamic global spatial prior for patch-level screening via joint-confidence scoring to suppress background-induced spurious responses, COFV performs SAR-aware feature routing to disentangle target-dominant, contextual, and noise-dominant scattering responses, and DEPART introduces module-specific hard-sample feedback to reinforce the scattering discrimination capacity of both COSD and COFV. Experiments on the FUSAR-Airport and FUSAR-Harbor datasets show that, across six evaluated baselines, CORE-Det reduces GFLOPs

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by 88.92% and inference time by 51.97% on average, while improving mAP_{50} by 9.59 pp and recall by 8.94 pp and reducing false alarms by 19.61 pp. These results indicate that aligning computation with scene structure, scattering responses, and sample difficulty is effective for large-scene SAR detection. Future work will focus on broader cross-sensor evaluation, stronger calibration under larger scene shifts, and validation on larger public benchmarks.

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